

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8        | (a) |      |  | <b>Advantages</b><br>award (1) each for up to <b>two</b> of following <ul style="list-style-type: none"> <li>cane sugar is a renewable raw material / sustainable</li> <li>plant photosynthesis uses CO<sub>2</sub> and fermentation produces CO<sub>2</sub> making it carbon neutral</li> <li>low pressure making it cheaper to operate / safer to operate</li> </ul><br><b>Disadvantage</b><br>award (1) each for up to <b>two</b> of following <ul style="list-style-type: none"> <li>dilute solution of ethanol formed / ethanol not pure - therefore needs further processing / distillation</li> <li>carbon dioxide formed - contributes to global warming</li> <li>batch process so labour intensive / inefficient / time consuming</li> <li>crops used therefore less land for food production / more expensive food</li> </ul> | 2               |     | 2   | 4     |       |      |
|          | (b) |      |  | $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$<br><b>both</b> products needed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 | 1   |     | 1     |       |      |
|          | (c) | (i)  |  | 498 (2)<br>if answer incorrect award (1) for either of following<br>5616 – (2 × 2061)<br>1494                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 | 2   |     | 2     | 2     |      |
|          |     | (ii) |  | 6932 (2)<br>if answer incorrect award (1) for (4 × 805) + (8 × 464)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 | 2   |     | 2     | 2     |      |